

NASHUA, NH, USA

WMO: 743946

Lat: 42.783N

Lon: 71.517W

Elev: 61

StdP: 100.59

Time Zone: -5.00 (NAE)

Period: 05-19

WBAN: 54754

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.4	-13.8	-24.0	0.4	-13.1	-21.9	0.5	-11.2	11.2	-4.0	10.1	-3.0	2.5	310	0.476

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.6	33.2	23.1	31.6	22.2	30.0	21.3	24.7	30.1	23.8	28.8	23.0	27.5	3.3	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.2	18.1	26.8	22.4	17.2	26.0	21.6	16.3	25.3	74.8	30.3	71.3	28.8	68.0	27.5	28.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.2	8.0	7.0	-21.1	36.0	3.5	1.4	-23.6	37.0	-25.7	37.8	-27.6	38.6	-30.2	39.6
			-21.6	26.2	3.2	1.0	-23.9	26.9	-25.8	27.5	-27.6	28.0	-29.9	28.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.9	-3.3	-2.6	2.1	8.8	15.2	19.6	23.3	21.8	17.6	11.2	4.7	-0.6		
	DBStd	10.33	5.63	5.13	5.24	4.79	4.50	3.75	2.94	2.98	4.15	4.31	4.69	5.07		
	HDD10.0	1641	414	355	252	78	6	0	0	0	0	37	171	329		
	HDD18.3	3489	672	587	503	288	121	28	2	5	63	225	410	586		
	CDD10.0	1599	1	1	7	43	167	289	412	367	229	73	10	1		
	CDD18.3	406	0	0	0	3	23	67	156	113	42	2	0	0		
	CDH23.3	4031	0	0	7	55	309	673	1546	1038	376	28	0	0		
	CDH26.7	1432	0	0	1	18	105	227	614	347	117	4	0	0		
(14) Wind	WSAvg	2.3	2.5	2.8	3.1	2.9	2.4	2.1	1.9	1.7	1.8	2.0	2.2	2.3		
(15) Precipitation	PrecAvg	1176	87	86	100	104	103	99	92	91	97	116	105	106		
	PrecMax	1493	153	292	183	234	296	278	153	231	232	362	180	177		
	PrecMin	833	26	42	18	14	21	23	37	4	32	10	34	33		
	PrecStd	177	36	52	40	62	60	68	32	58	55	87	39	39		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.8	16.3	23.0	29.3	32.8	34.5	35.4	34.1	32.9	26.7	20.5	16.5		
		MCWB	13.1	10.9	13.0	16.4	20.2	23.4	24.5	24.0	23.4	18.9	15.4	13.9		
	2%	DB	10.9	10.7	17.2	24.1	29.8	31.5	33.3	32.0	29.9	23.4	17.7	12.0		
		MCWB	7.7	5.8	11.1	14.4	18.7	21.9	23.4	22.7	21.7	18.2	13.3	9.5		
	5%	DB	7.0	7.3	13.1	20.8	26.8	29.2	31.8	30.1	27.6	20.9	15.3	9.2		
		MCWB	4.3	3.7	7.1	12.6	17.6	20.2	22.6	21.8	20.8	17.3	11.5	6.4		
	10%	DB	4.3	4.5	10.2	18.0	24.1	27.3	30.3	28.5	25.3	18.4	12.8	6.8		
		MCWB	1.6	1.6	5.6	11.3	16.3	19.0	21.8	20.7	19.3	14.4	9.7	4.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.8	13.1	15.3	18.9	22.1	24.5	26.3	25.9	24.7	21.5	17.5	14.1		
		MCDB	14.4	15.2	21.3	24.1	30.4	31.6	32.0	32.4	29.5	23.7	19.6	15.5		
	2%	WB	8.0	7.0	11.9	16.5	20.6	23.3	24.9	24.5	23.2	19.7	14.6	10.3		
		MCDB	9.9	10.2	15.8	22.1	26.5	29.4	30.6	29.1	27.8	21.5	16.4	11.6		
	5%	WB	4.4	4.1	8.4	14.1	19.3	22.1	24.0	23.5	22.2	17.5	12.4	7.1		
		MCDB	6.7	6.7	11.7	18.7	24.4	27.2	29.2	27.4	25.9	20.2	14.4	8.5		
	10%	WB	2.0	2.1	5.9	11.6	17.6	20.8	23.1	22.4	20.8	15.6	10.2	4.6		
		MCDB	3.9	4.0	9.6	16.7	22.3	24.3	27.9	26.4	23.8	17.8	12.7	6.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.6	10.0	10.4	11.8	12.1	11.3	11.6	11.7	11.8	11.1	10.1	8.8		
		MCDBR	12.0	14.0	15.1	16.9	16.5	14.1	13.5	13.3	14.0	13.6	14.0	12.0		
	5% WB	MCWBR	10.1	10.2	9.4	8.8	7.5	6.1	5.4	5.7	7.0	8.5	10.0	10.1		
		MCWBR	11.8	12.8	13.4	14.6	14.1	12.2	11.5	11.0	11.5	11.5	11.2	11.2		
(40) Clear-Sky Solar Irradiance	taub		0.284	0.295	0.322	0.363	0.399	0.439	0.460	0.427	0.373	0.332	0.307	0.287		
	taud		2.458	2.438	2.451	2.388	2.326	2.241	2.192	2.266	2.398	2.528	2.553	2.541		
	Ebn at Noon		864	912	924	904	875	838	814	831	856	859	833	833		
	Edh at Noon		73	88	99	114	125	137	142	128	104	81	68	63		
(44) All-Sky Solar Radiation	RadAvg		1.78	2.64	3.65	4.62	5.08	5.50	5.81	5.24	4.24	2.74	1.89	1.44		
	RadStd		0.11	0.20	0.24	0.50	0.61	0.53	0.36	0.27	0.29	0.26	0.16	0.15		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (78 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	+90	+36

Nomenclature: See separate page